NOTE: The document identifier and heading has been changed on this page to reflect that this is a performance specification. There are no other changes to this document. The document identifier on subsequent pages has not been changed, but will be changed the next time this document is revised.

MIL-PRF-6106/43A(USAF)
29 April 1985
SUPERSEDING
MIL-R-6106/43(USAF)
14 October 1983

PERFORMANCE SPECIFICATION SHEET

RELAYS, ELECTROMAGNETIC, 65 AMPERES, 1 PNO DOUBLE BREAK, HERMETICALLY SEALED, TYPE I

This specification is approved for use by the Department of the Air Force, and is available for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the relays described herein shall consist of this specification and the latest issue of MIL-PRF-6106.

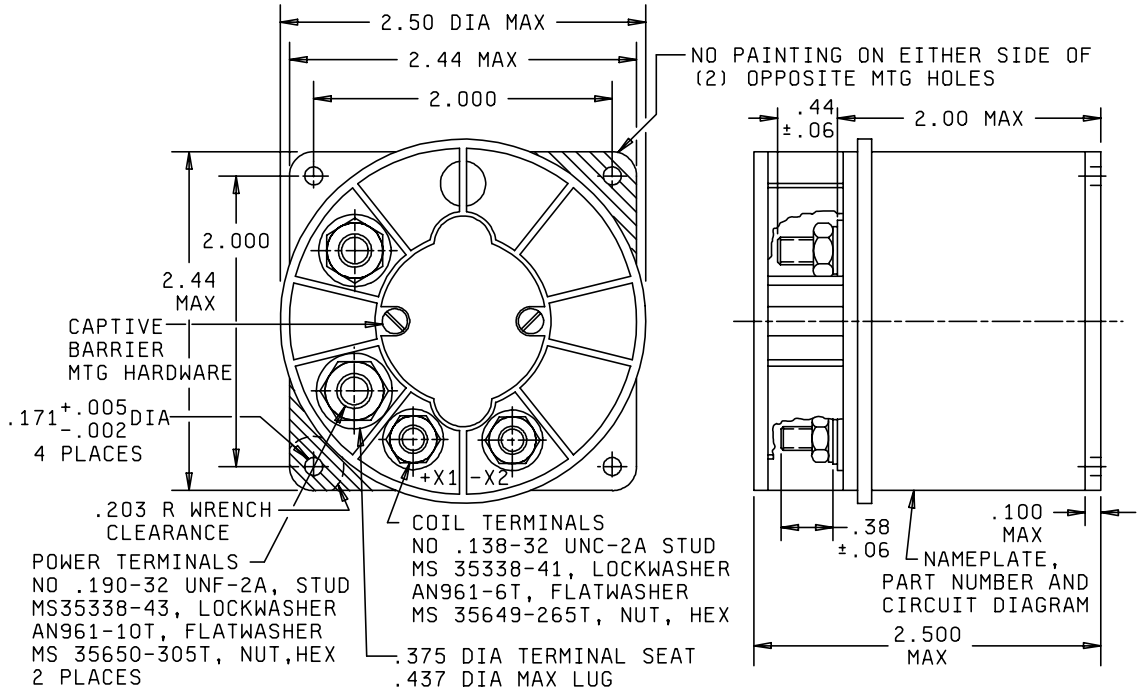
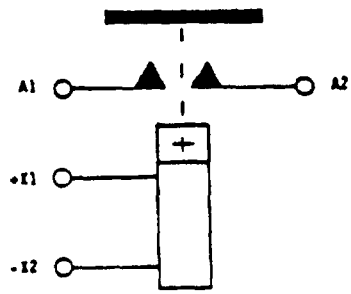


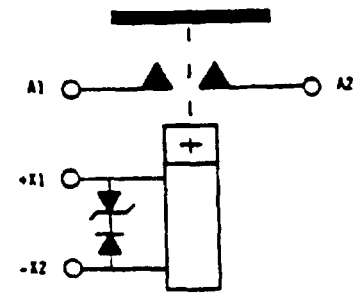
FIGURE 1. Relay, outline drawing.

FSC 5945

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TYPE A



TYPE B

OBSERVE POLARITY, DC COILS
CIRCUIT DIAGRAM

INCHES	MM	INCHES	MM	INCHES	MM
.005	0.13	.203	5.16	1.910	48.51
.010	0.25	.375	9.53	2.000	50.80
.060	1.52	.380	9.65	2.440	61.98
.100	2.54	.437	11.11	2.500	63.50
.171	4.34	.440	11.18		

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerance is ±.010 (0.25 mm).
4. Terminal numbers need not appear on the relay header provided there is affixed to the relay a suitable legible circuit diagram that identifies each terminal location specified.

FIGURE 1. Relay, outline drawing - Continued.

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TABLE I. Operating characteristics

Dash No	Coil data													Time-milliseconds-max 2/					
	Coil	Nominal			Max		Max pick-up voltage			Drop out voltage 1/	Hold voltage 1/	Coil sup. back EMF max	CKT. type	Operate 2/	Release 3/	Bounce			
		Volts	Freq. Hz	Res Ω ± 10%	Volts	Amp @ 25°C	Nominal 1/	High temp test	Cont. current test							Main		Aux.	
																NO	NC	NO	NC
-001	X1X2	28	DC	200	29	.160	18	20	21	1.5	7.0	--	A	35	25	3	--	5	--
-002	X1X2	28	DC	200	29	.160	18	20	21	1.5	7.0	50	B	35	25	3	--	5	--

1/ Over temperature range.
2/ With nominal coil voltage.
3/ From nominal coil voltage.

TABLE II. Rated contact load (amperes per pole) case grounded.

Type of load	Life operating cycles X 10^3	28 V dc				115 V ac, 1 phase				115/200 V ac, 3 phase 1/				See appropriate notes
		Main		Aux		Main		Aux		Main		Aux		
		NO	NC	NO	NC	400 Hz	50/60 Hz	400 Hz	60 Hz	400 Hz	50/60 Hz	400 Hz	60 Hz	
Resistive	50	50	---	---	---	65	---	---	---	---	---	---	---	
Inductive	20	50	---	---	---	---	---	---	---	---	---	---	---	
Inductive	50	---	---	---	---	65	---	---	---	---	---	---	---	
Lamp	50	17	---	---	---	---	---	---	---	---	---	---	---	
Lamp	25	20	---	---	---	---	---	---	---	---	---	---	---	
Transfer load		---	---	---	---	---	---	---	---	---	---	---	---	
Mechanical life reduced current	200	7	---	---	---	---	---	---	---	---	---	---	---	
Min. current	50	2.5	---	---	---	5	---	---	---	---	---	---	---	

1/ Absence of value indicates relay is not rated for 3 phase.

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REQUIREMENTS:

Coil data.

Coil data: See table I.

Duty rating: Continuous.

Coil suppression, back EMF. See table I.

Operational data:

Operate time: See table I.

Release time: See table I.

Rated contact load See table II.

Physical data:

Dimensions and configuration See figure 1.

Weight: 0.688 pound (312 grams) max.

Seal: Hermetic.

Finish: Flat black.

Terminals: See figure 1.

Terminal insulator, terminal barrier: Melamine fiberglass, per MIL-P-15037, nylon 6/6 type, grade A, class 1, black per federal specification LP-395.

Environmental characteristics:

Temperature range -55°C to +71°C.

Altitude: 50,000 feet, maximum.

Vibration - sinusoidal:

Amplitude, double. 06 inch.

Frequency range: 10 - 55 Hz.

G-level: 10 G's.

(A) Frequency range: 55 - 2000 Hz.

(A) Vibration - Random:

1 hour each plane

0.04 G^2/Hz 15-197 cycles

4 dB/octave increase

197-300 cycles

0.07 G^2/Hz 300-1000 cycles

6 dB octave decrease

1000-2000 cycles

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Shock G level. 25 G's.

Duration: 6 - 9 milliseconds.

Maximum duration contact opening: 10 microseconds.

Acceleration. 10 G's

Electrical:

Insulation resistance, initial: 100 megohms.

After life or environmental tests. 50 megohms.

Contact voltage drop, initial· 0.150 volts maximum.

After life test: 0.175 volts maximum.

Overload current: 125 amperes dc, 400 amperes ac.

Rupture current. 150 amperes dc, 500 amperes ac.

RFI standard (applicable to coil currents of ac operated relays): MIL-STD-461.

Dielectric strength (sea level).

	Initial	After life tests
Coil to case	1000 V rms	1000 V rms
Auxiliary contacts	1000 V rms	750 V rms
All other points	1500 V rms	1150 V rms

Dielectric strength at 50,000 ft. altitude

Coil to case	500 V rms
Auxiliary contacts	
All other points	700 V rms

Part number M6106/43 (dash number from table I).

Custodian:
Air Force - 85

Review activities·
Air Force - 99
DLA - ES

Agent·
DLA - ES

Preparing activity.
Air Force - 85

(Project 5945-F596-2)

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